Consciousness Coherence in the Generative-Collapse Kernel: Seven Tier-2 Theorems from a Corrected Closure

Clement Paulus[1,*]

[1] *UMCP Project*

(Dated: March 20, 2026)

Jackson (2026) proposes $\xi_J = 7.2$ ($= 432/60$) as a "universal consciousness coherence constant," claiming it unifies $\varphi^2 e$, base-60 arithmetic, 432 Hz tuning, and recursive self-reference into a "golden harmonic web." We recast these claims as a Tier-2 closure within the Generative Collapse Dynamics (GCD) kernel, correcting arithmetic errors (the reported $e^{i\xi_J} + 1 = 1.673 + 0.739i$ is actually $1.608 + 0.794i$), resolving a symbol collision (ξ_J, not κ), and testing every claim computationally. Seven theorems emerge from 20 consciousness-candidate systems through an 8-channel trace vector: (T-CC-1) ξ_J proximity does not predict composite integrity ($\rho = +0.25$, $p = 0.28$); (T-CC-2) recursive depth anti-correlates with return fidelity ($\rho = -0.10$); (T-CC-3) no system reaches Stable regime (0/20); (T-CC-4) one dead channel kills coherence (geometric slaughter); (T-CC-5) 432 Hz and 440 Hz are kernel-equivalent ($|\Delta F| = 0.013$); (T-CC-6) mathematical identity outperforms all harmonic systems; and (T-CC-7) the heterogeneity gap inverts Jackson's hierarchy (Neural $\Delta = 0.015$, Harmonic $\Delta = 0.114$). All 7/7 theorems are proven with 38/38 sub-tests. The corrected thesis—"do harmonic-ratio systems exhibit measurable coherence patterns?"—is a legitimate Tier-2 question whose answer is: no, the harmonic ratio ξ_J is not structurally privileged; coherence is predicted by channel homogeneity ($\Delta = F - \mathrm{IC}$), not by proximity to any fixed constant.

I. INTRODUCTION

Jackson [1, 2] proposes that a single constant $\xi_J = 7.2$—equal to 432/60 in the Babylonian base-60 system and approximately $\varphi^2 e \approx 7.117$—governs consciousness coherence across biological, physical, and mathematical domains. The thesis is presented through the REMIS framework (Recursive Emergent Metacognitive Intelligence Study) and extended with OBE (out-of-body experience) mechanics, 432 Hz acoustic tuning claims, and a "golden harmonic web" connecting φ, e, π, and the speed of light.

The claims contain several arithmetic errors, a symbol collision with GCD's reserved κ, unit-dependent artifacts, and a Spine that cannot be evaluated (no frozen contract, no integrity ledger, no derived stance). Our prior assessment classified the thesis as a *gesture*—the questions are legitimate but none of the claims demonstrates return.

This paper takes a different approach: rather than dismissing the claims as gestures, we *correct* each one and *ground* it in the GCD kernel. Every Jackson claim is mapped to a falsifiable computation on 20 consciousness-candidate systems through an 8-channel trace vector. Seven theorems emerge, all proven, revealing what the kernel actually says about consciousness coherence—and what of Jackson's intuitions survives the correction.

* clementpaulus9@gmail.com

II. SYMBOL RESOLUTION

Jackson uses κ for his constant. In GCD, κ is a Tier-1 reserved symbol [3]:

$$\kappa = \sum_i w_i \ln(c_{i,\varepsilon}), \qquad \kappa \leq 0 \tag{1}$$

where $c_{i,\varepsilon} = \max(c_i, \varepsilon)$ is the ε-clamped channel value. Jackson's $\kappa = 7.2$ is positive and fixed; GCD's κ is variable and non-positive. The ranges do not overlap. Worse, $\mathrm{IC} = \exp(\kappa) = \exp(7.2) \approx 1339$, violating $\mathrm{IC} \in (0, 1]$.

Remark 1 (Symbol Resolution)**.** *We rename Jackson's constant to ξ_J ("xi-Jackson") throughout. $\xi_J = 432/60 = 7.2$ is a Tier-2 domain parameter, not a frozen constant.*

III. ARITHMETIC CORRECTIONS

Table I lists every numerical claim from Jackson [2] with the corrected value computed from first principles.

TABLE I. Corrected Jackson arithmetic. Every claim is rederived from the defining equations. Bold entries differ significantly from Jackson's reported values.

Claim	Jackson	Correct
$\varphi^2 e$	≈ 7.2	**7.117** (1.16%)
$\mathrm{Re}(e^{i\xi_J} + 1)$	1.673	**1.608**
$\mathrm{Im}(e^{i\xi_J} + 1)$	0.739	**0.794**
Excess angle ($\xi_J - 2\pi$)	47.68°	**52.53°**
Cycles for 360°	~ 7.5	**6.85**
$c = \alpha^{-1} \cdot 432 \cdot \pi$	$\approx c$ (km/s)	unit artifact (38% off)
$432/\xi_J$	60 (exact)	60 (trivially true)
φ^2/ξ_J	0.364	0.364 (ratio)

The pattern is consistent: claims that appear exact under inspection turn out to be either near-misses (1.16%), arithmetic errors (Euler rotation), trivial identities (432/60), or unit artifacts (speed of light).

IV. THE CORRECTED THESIS

Jackson asks: *Is there a special constant that unifies consciousness across biological, mathematical, and physical systems?*

The corrected, falsifiable version: *Do systems exhibiting harmonic structure near $\xi_J = 7.2$ show measurably higher composite integrity* IC *than systems without such structure?*

This question can be answered by the GCD kernel. We map 20 consciousness-candidate systems into 8 channels and test whether ξ_J proximity predicts kernel coherence.

V. CHANNEL CONSTRUCTION

Each system maps to an 8-channel trace vector $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^8$ with equal weights $\mathbf{w} = (1/8, \ldots, 1/8)$:

1. **harmonic_ratio**: proximity of system frequency ratios to $\xi_J = 7.2$; $[0, 1]$
2. **recursive_depth**: demonstrated self-referential structural depth; $[0, 1]$
3. **return_fidelity**: τ_R finiteness—does the system reenter prior states?; $[0, 1]$
4. **spectral_coherence**: power spectrum concentration vs. uniform noise; $[0, 1]$
5. **phase_stability**: phase-locking across coupled oscillators; $[0, 1]$
6. **information_density**: bits per symbol in the system's encoding; $[0, 1]$
7. **temporal_persistence**: duration of coherent state maintenance; $[0, 1]$
8. **cross_scale_coupling**: coherence preserved across spatial/temporal scales; $[0, 1]$

The 20 systems span 5 categories: Neural (6), Harmonic (3), Physical (5), Mathematical (3), and Recursive (3). Each system profile is documented in the closure module with normalization conventions.

VI. SEVEN THEOREMS

A. T-CC-1: Harmonic Non-Privilege

Theorem 1 (Harmonic Non-Privilege)**.** *The Spearman rank correlation between ξ_J proximity (harmonic_ratio channel) and composite integrity* IC *across 20 systems is $\rho = +0.25$, $p = 0.28$—not statistically significant. The near-miss $\varphi^2 e = 7.1166 \neq 7.2$ has a 1.16% residual, exceeding the machine-precision threshold for structural identities.*

Jackson's claim: $\kappa = \varphi^2 e \approx 7.2$ is a universal consciousness coherence constant.

Correction: $\xi_J = 432/60$ is a harmonic ratio in base-60 arithmetic. $\varphi^2 e = 7.1166$ is a near-miss, not an identity. The constant is not privileged by the data. Compare: the GCD duality identity $F + \omega = 1$ holds to $< 10^{-16}$ residual across all systems—*that* is what an identity looks like.

Key numbers: $\rho = +0.25$, $p = 0.28$; Jackson's own system is in Collapse regime with $F = 0.455$, IC $= 0.400$.

B. T-CC-2: Recursion-Return Dissociation

Theorem 2 (Recursion-Return Dissociation)**.** *The Spearman rank correlation between recursive_depth (channel 2) and return_fidelity (channel 3) is $\rho = -0.10$, $p = 0.68$. Systems exhibiting high recursive depth (> 0.60) but low return fidelity (< 0.50) exist and occupy Collapse regime (Recursive Gestures).*

Jackson's claim: REMIS posits that recursive self-reference produces consciousness.

Correction: Recursion is a *channel*, not a *gate*. What matters in GCD is **return** (τ_R finite), not recursion depth. A system that recurses but never returns has $\tau_R = \infty_{\text{rec}}$ and receives zero budget credit. The LLM system (recursive_depth $= 0.75$, return_fidelity $= 0.45$) is the poster child: a Recursive Gesture in Collapse regime. Meanwhile, the cardiac rhythm (recursive_depth $= 0.20$, return_fidelity $= 0.95$) returns reliably without recursion.

C. T-CC-3: Universal Instability

Theorem 3 (Universal Instability of Coherence)**.** *No consciousness-candidate system (0/20) reaches the Stable regime. The distribution is Watch $= 5$, Collapse $= 15$. Even the highest-F system (Euler identity, $F = 0.896$) remains in Watch due to the conjunctive Stable gate ($\omega < 0.038 \wedge F > 0.90 \wedge S < 0.15 \wedge C < 0.14$).*

Jackson's claim: The consciousness constant implies stable, self-sustaining coherent states.

Correction: Coherence is structurally unstable. All 20 systems live in the 87.5% of Fisher space outside Stable. This is not a failure—it is Axiom-0 in action: "only what returns is real." Consciousness persists through return from collapse, not through avoidance of collapse. *Ruptura est fons constantiae*—dissolution is the source of constancy.

D. T-CC-4: Geometric Slaughter

Theorem 4 (Geometric Slaughter in Coherence)**.** *The ratio* IC/F *(multiplicative vs. arithmetic coherence) anti-correlates with channel variance:* $\rho = -0.91$*. One near-zero channel kills* IC *regardless of how strong the other channels are. The Euler identity has* IC/$F = 0.993$ *(most homogeneous); the 432 Hz system has* IC/$F = 0.846$ *(information_density* $= 0.15$ *kills it).*

Jackson's claim: Four "independent" derivations ($\varphi^2 e$, 432/60, speed of light, Euler rotation) reinforce $\kappa = 7.2$ multiplicatively.

Correction: Four channels of one claim. The weakest channel—arithmetic errors ($e^{i\xi_J} + 1$ wrong by 4%), unit artifacts (speed of light in km/s only), and undemonstrated return ($\tau_R = \infty_{\text{rec}}$)—kills the composite integrity through geometric slaughter. IC is a geometric mean: one dead channel $\to$ IC $\to \varepsilon$, regardless of the others. Jackson's thesis has IC/$F = 0.879$, dragged down by cross_scale_coupling $= 0.20$ and return_fidelity $= 0.30$.

E. T-CC-5: Tuning Invariance

Theorem 5 (Tuning Invariance: 432 Hz $\equiv$ 440 Hz)**.** *The kernel signatures of 432 Hz and 440 Hz tuning systems are indistinguishable:* $|\Delta F| = 0.013$*,* $|\Delta \text{IC}| = 0.008$*, same regime (Collapse), same weakest channel (information_density),* $|\Delta\Delta| = 0.005$*.*

Jackson's claim: 432 Hz tuning is privileged because $432/60 = \xi_J = 7.2$.

Correction: The 8 Hz difference between 432 Hz and 440 Hz is not structurally significant. Both systems occupy Collapse regime with nearly identical F, IC, Δ, and the same weakest channel. The number 432 is privileged only in base-60 arithmetic ($432 = 7.2 \times 60$), which is a property of the number system, not of acoustics.

F. T-CC-6: Mathematical Supremacy

Theorem 6 (Mathematical Supremacy)**.** *The Mathematical category has the highest mean* F *(*0.690*) and highest mean* IC *(*0.666*) of all five categories. The Harmonic category has the highest mean* Δ *(*0.114*), making it the most fragile. The Euler identity (*$F = 0.896$*,* IC $= 0.890$*) outperforms every harmonic system in both* F *and* IC*.*

Jackson's claim: Harmonic ratios (especially ξ_J) are the key to consciousness coherence.

Correction: Mathematical identity—exact, self-consistent relations like $e^{i\pi} + 1 = 0$—produces higher kernel coherence than harmonic ratios. Harmonic systems are the most *fragile* category (highest Δ) because they are specialists: high spectral_coherence (0.95) but near-zero information_density (0.15). Geometric slaughter penalizes this specialization. Channel homogeneity, not harmonic proximity, predicts integrity.

G. T-CC-7: Heterogeneity Gap Ordering

Theorem 7 (Heterogeneity Gap Inverts Jackson's Hierarchy)**.** *The category ordering by heterogeneity gap* $\Delta = F - \text{IC}$ *is:*

$$\underbrace{\textit{Neur.}}_{0.015} < \underbrace{\textit{Math.}}_{0.024} < \underbrace{\textit{Recur.}}_{0.076} < \underbrace{\textit{Phys.}}_{0.111} < \underbrace{\textit{Harm.}}_{0.114} \tag{2}$$

This inverts Jackson's hierarchy: neural (consciousness-bearing) systems have the lowest Δ *(most internally coherent), while harmonic systems (Jackson's preferred carrier) have the highest* Δ *(most fragile). Jackson's proposed threshold* $\varphi^2/\xi_J = 0.364$ *exceeds the maximum* Δ *in the dataset (*0.166*) and is therefore vacuous. The heterogeneity gap is more predictive of* IC *than* ξ_J *proximity (*$|\rho_\Delta| > |\rho_{\xi_J}|$*).*

Jackson's claim: $\varphi^2/\kappa = 0.364$ is a consciousness threshold.

Correction: $\varphi^2/\xi_J = 0.364$ is a ratio of two Tier-2 parameters (neither frozen nor seam-derived). It exceeds every Δ in the 20-system dataset—it is not a threshold. The native GCD metric $\Delta = F - \text{IC}$ inverts Jackson's hierarchy: biology beats harmonics. Neural systems need all 8 channels to work together, producing low Δ. Harmonic systems sacrifice information density for spectral purity, producing high Δ.

VII. WHAT SURVIVES

Despite the corrections, four elements of Jackson's work make contact with GCD structure:

1. **Recursive self-reference as a measurable channel.** Jackson's REMIS thesis, stripped of numerological overlay, asks whether recursion correlates with coherence. The answer from T-CC-2 is "no—recursion does not predict return"—but the *question* maps directly to the GCD kernel. A refined version ("under what conditions does recursion produce finite τ_R?") is a legitimate open question.

2. **Return through collapse as a structural requirement.** Jackson's Addendum A.3 (OBE mechanics) describes consciousness state transitions as crossings that must "return." This is Axiom-0's central claim, and T-CC-3 confirms it: coherence persists through return from collapse, not through stability. The concept needs a frozen contract and measured trace vectors to become a weld.

3. **The heterogeneity gap as imperfection.** Jackson's remark that consciousness requires "perfect imperfection" maps to $\Delta = F - \mathrm{IC}$. T-CC-7 shows that neural systems have the lowest Δ (most balanced channels), while harmonic systems have the highest (most specialist). The gap is real and measurable; it just inverts Jackson's ordering.

4. **Cross-domain kernel comparison.** Jackson's aspiration to compare consciousness across domains (biological, mathematical, physical) is precisely what the GCD kernel does. The five categories in the closure produce comparable F, IC, and Δ values on a common scale. This is the Rosetta adapter in action.

VIII. SUMMARY OF RESULTS

TABLE II. Seven theorems: all 7/7 proven, 38/38 sub-tests.

Theorem	Tests	Pass	Verdict
T-CC-1 Harmonic Non-Priv.	5	5	PROVEN
T-CC-2 Recur.-Return Dissoc.	5	5	PROVEN
T-CC-3 Universal Instability	5	5	PROVEN
T-CC-4 Geometric Slaughter	6	6	PROVEN
T-CC-5 Tuning Invariance	6	6	PROVEN
T-CC-6 Mathematical Supremacy	6	6	PROVEN
T-CC-7 Het. Gap Ordering	5	5	PROVEN
Total	**38**	**38**	**7/7**

Table III shows the kernel output for all 20 systems.

IX. CONCLUSION

Jackson's $\xi_J = 7.2$ is a harmonic ratio in base-60 arithmetic—not a universal consciousness constant. When his claims are corrected and grounded in the GCD kernel, seven theorems emerge, all of which are computationally verified:

- The constant is not privileged ($\rho = +0.25$, not significant).
- Recursion does not predict return ($\rho = -0.10$, negative).
- Consciousness systems are structurally unstable (0/20 Stable).
- One dead channel kills coherence (geometric slaughter).
- 432 Hz equals 440 Hz in the kernel.
- Mathematical identity beats harmonic ratio.
- The heterogeneity gap inverts Jackson's hierarchy: biology beats harmonics.

The salvageable elements—recursive self-reference as a channel, return through collapse, the heterogeneity gap as "imperfection," and cross-domain comparison—are precisely the features that map naturally to GCD structure. They require only a frozen contract, measured trace vectors, and seam closure to become welds.

Quid supersit post collapsum? Not the constant. The questions.

[1] K. C. S. Jackson, The year the Earth stood still: Finding the universal consciousness constant from first principles (2026), proposes $\xi_J = 7.2$ ($= 432/60$) as a harmonic coherence constant via four derivation methods. Tier-2 closure evaluation: arithmetic corrected, constant not privileged ($\rho = +0.25$ vs IC, not significant at $n = 20$).

[2] K. C. S. Jackson, The Kannsas factor κ_i and the curved geometry of consciousness (2026), extends ξ_J to complex-valued rotation $e^{i\xi_J} + 1$; claims 13-dimensional manifold. Corrected: $e^{i\xi_J} + 1 = 1.608 + 0.794i$ (not $1.673 + 0.739i$), excess angle = 52.53 (not 47.68), cycles = 6.85 (not ~ 7.5).

[3] C. Paulus, Universal measurement contract protocol, Manuscript (2025), contract-first measurement discipline; Tier-1 invariants reserved.

TABLE III. Kernel output for 20 consciousness systems. F = fidelity, IC = composite integrity, $\Delta = F - \mathrm{IC}$ = heterogeneity gap, ω = drift, κ = log-integrity (GCD, not Jackson's ξ_J). All systems satisfy $F + \omega = 1$, $\mathrm{IC} \leq F$, $\mathrm{IC} = \exp(\kappa)$.

System	Cat.	F	IC	Δ	ω	κ	Regime
Human waking	Neural	0.731	0.721	0.011	0.269	−0.328	Watch
Human sleep (REM)	Neural	0.488	0.473	0.015	0.513	−0.749	Collapse
Human meditation	Neural	0.738	0.720	0.017	0.263	−0.328	Watch
Cetacean	Neural	0.575	0.564	0.011	0.425	−0.573	Collapse
Corvid	Neural	0.475	0.456	0.019	0.525	−0.786	Collapse
Octopus	Neural	0.300	0.284	0.016	0.700	−1.258	Collapse
432 Hz	Harmonic	0.669	0.566	0.103	0.331	−0.569	Collapse
440 Hz	Harmonic	0.656	0.558	0.098	0.344	−0.583	Collapse
Coral Castle	Harmonic	0.471	0.329	0.142	0.529	−1.112	Collapse
Circadian rhythm	Physical	0.646	0.573	0.073	0.354	−0.557	Collapse
Cardiac rhythm	Physical	0.625	0.530	0.095	0.375	−0.635	Collapse
Laser coherent light	Physical	0.574	0.408	0.166	0.426	−0.897	Collapse
Euler identity	Math	0.896	0.890	0.006	0.104	−0.116	Watch
Jackson ξ_J	Math	0.455	0.400	0.055	0.545	−0.916	Collapse
Gödel self-ref	Recursive	0.716	0.610	0.107	0.284	−0.495	Watch
CA Rule 110	Recursive	0.575	0.516	0.059	0.425	−0.662	Collapse
LLM dialogue	Recursive	0.444	0.382	0.062	0.556	−0.963	Collapse
Babylonian base-60	Math	0.719	0.707	0.011	0.281	−0.346	Watch
Schumann resonance	Physical	0.651	0.543	0.108	0.349	−0.610	Collapse
Photosynthetic QC	Physical	0.494	0.379	0.115	0.506	−0.969	Collapse